

NSRC: EVENT-ADDRESSED RESIDUAL MEMORY FOR RELIABLE NEURAL FORECAST CORRECTION

Litong Sun¹ Peizong Yang¹ YangTian Mi¹ Xiapu Luo² Weigang Wu^{1*}

¹ Sun Yat-sen University; ² Hong Kong Polytechnic University, Hong Kong SAR
¹ {sunlt5, yangpz3, miyt}@mail2.sysu.edu.cn; wuweig@mail.sysu.edu.cn

ABSTRACT

Event announcements can induce delayed, structured deviations from price-only forecasts. We introduce Neural Structured Residual Correction (NSRC), which stores mature forecast errors with explicit event addresses and retrieves them as correction trajectories. Semantic matching, volatility-aware transfer, delayed-feedback calibration, and a validation-frozen router adapt the correction while preserving a numerical forecast anchor. Across celebrity events, general news, cryptocurrency macro announcements, and equity-index announcements, NSRC reduces equal-cohort DLinear MSE/MAE/MAPE by 16.60%/13.44%/13.36%, leads learned baselines in 11 of 12 cohort-metric cells, and has eight exploratory NSRC-vs.-DLinear contrasts that remain significant after Holm correction. Because the test intervals also informed method exploration, we report retrospective within-period results.

Index Terms— Event-conditioned forecasting, structured residual memory, time-series forecasting, residual correction

1. INTRODUCTION

Public information and social narratives can shape asset prices, but event responses differ across assets and horizons; event studies and social-media analyses document this setting [1, 2, 3, 4]. A price history alone cannot explain all multi-step deviations. As Fig. 1 shows, event classes induce distinct residual distributions and temporal patterns, motivating reuse of event-conditioned evidence rather than pooling these errors as undifferentiated noise.

NSRC keeps the numerical forecaster fixed and reuses its mature event-conditioned errors. Each completed forecast stores an address (c, d, z) —category, direction, and description—with its residual trajectory. For a new event, regularized retrieval and volatility-aware transfer adapt related trajectories, delayed-feedback calibration sets their strength, and a validation-frozen router selects the deployed path.

Financial forecasting has incorporated social-media signals, multimodal price/text features, and event-aware transformers [5, 6, 7]; recent time-series foundation models also align numerical and textual streams [8, 9]. RATL stores a frozen forecaster’s residual trajectories and retrieves them by numerical context, with routing over forecast blocks and variables [10]. NSRC complements this residual-memory line by addressing residuals with event category, direction, and description rather than numerical context alone, then aligning transfer by volatility and calibrating it with delayed feedback. Thus, recurring catalysts—not only similar price windows—organize residual reuse. Inspired by inspectable structured and neuro-symbolic representations [11, 12], NSRC keeps the event state explicit without requiring

a symbolic program; Table 1 compares it with forecasting baselines. Our contributions are:

1) Event-conditioned residual formulation. We represent each completed forecast’s mature multi-step error as an event-conditioned residual trajectory, admitting it to memory only after the full target horizon is observed.

2) Semantic residual memory and calibrated transfer. A category–direction–description address retrieves related trajectories; recency and asset weights, volatility alignment, and support shrinkage govern transfer, while delayed feedback calibrates its amplitude.

3) Evidence-guided deployment. A validation-frozen, cohort-level router selects among semantic and uniform-memory corrections, the numerical backbone, and persistence; across four cohorts it determines when to retain an anchor or deploy a calibrated blend. Code is open-source at github.com/SeamewZ/NSRC.

2. NEURAL STRUCTURED RESIDUAL CORRECTION

Fig. 1 summarizes the NSRC pipeline. At each event origin, a frozen numerical backbone issues an H -step forecast. Once historical outcomes mature, NSRC stores their residual trajectories with event addresses; a new event retrieves related records, adapts their scale, and calibrates the correction using only earlier outcomes. A validation-frozen router selects among NSRC’s correction paths and forecasting anchors.

Forecasting interface. A frozen backbone maps the last $L = 96$ observations to $\hat{\mathbf{u}}_t^b = f_\theta(\mathbf{X}_t)$, where $u_{t,h} = P_{t+h}/P_t$ for $h = 1, \dots, H$. We use $H = 24$ hourly steps for crypto and 24 valid trading observations for equities. Event timestamps are floored to the hour and aligned one hour earlier, as in the experimental protocol; corrections are applied after release, and events arriving after a forecast window begins are excluded from that forecast. The residual $\delta_t = \mathbf{u}_t - \hat{\mathbf{u}}_t^b$ enters memory only after its full target horizon has matured.

2.1. Semantic addressing of residual trajectories

The event address $e = (c, d, z)$ records event category, direction, and description. For celebrity/news, GPT, Gemini, and DeepSeek independently assign the target-asset direction and majority voting yields d ; for macro events, d is determined by the released value relative to consensus. We trim and lowercase c, d . For description matching, T lowercases z , tokenizes alphanumeric strings, and removes single-character tokens and fixed stop words; Jaccard overlap of the resulting token sets lets differently worded descriptions share memory when they retain common terms.

Relevance weights. The causal archive $\mathcal{M}_t$ contains event origins from the preceding 90 calendar days whose full target horizons

*Corresponding author: Weigang Wu (wuweig@mail.sysu.edu.cn).

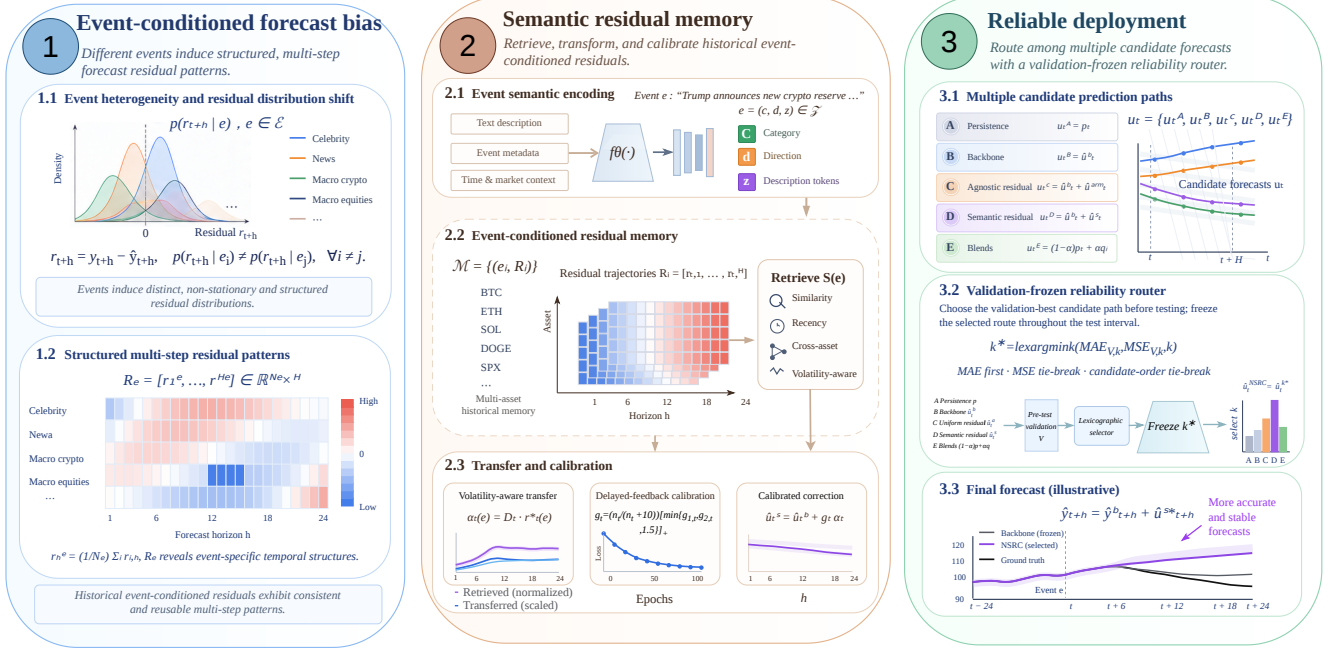


Fig. 1. NSRC pipeline: event-conditioned residual structure, semantic memory with calibrated transfer, and validation-frozen forecast routing.

are observed before t . We score agreement between a query event and each event stored in memory as

$$s(e, e') = \frac{1}{2} \mathbf{1}[c = c'] + \frac{1}{4} \mathbf{1}[d = d'] + \frac{1}{4} J(T(z), T(z')). \quad (1)$$

For historical origin i , let $\mathcal{E}_i$ be its associated events and $q_{ti}(e) = \max_{e' \in \mathcal{E}_i} s(e, e')$. Weights combine the best event match, a 60-day recency half-life, and asset proximity:

$$\tilde{q}_{ti}(e) = \begin{cases} q_{ti}(e), & \text{if } \exists j \in \mathcal{M}_t : q_{tj}(e) > 0, \\ 1, & \text{otherwise,} \end{cases} \quad (2)$$

$$w_{ti}(e) = \rho_{ti} 2^{-\Delta_{ti}/60} \tilde{q}_{ti}(e),$$

Here Δ_{ti} is age in days and $\rho_{ti} = 1$ for the same asset or .25 otherwise. Description matching is lexical token overlap, not embedding similarity. If any archive record has positive match score, zero-match records receive zero weight; if none match, all records fall back to recency and asset weights.

Volatility-aligned memory. For each record, input-window return volatility σ_i (floored at 10^{-4}) and the horizon index define a scale matrix. We normalize and clip residuals as follows:

$$\mathbf{D}_i = \sigma_i \text{diag}(1, \sqrt{2}, \dots, \sqrt{H}), \quad \mathbf{v}_i = \text{clip}(\mathbf{D}_i^{-1} \delta_i, -5, 5). \quad (3)$$

Stacking $\mathbf{v}_i^\top$ gives $\mathbf{V}_t \in \mathbb{R}^{|\mathcal{M}_t| \times H}$. Clipping is elementwise to $[-5, 5]$; the shared volatility–horizon coordinates support cross-asset transfer.

Regularized residual readout. For each query, let $\mathbf{w}_t(e) = [w_{ti}(e)]_{i \in \mathcal{M}_t}$. With $\lambda = 10$, NSRC computes:

$$\mathbf{r}_t^*(e) = \frac{\mathbf{V}_t^\top \mathbf{w}_t(e)}{\lambda + \mathbf{1}^\top \mathbf{w}_t(e)}, \quad \mathbf{a}_t(e) = \mathbf{D}_t \mathbf{r}_t^*(e). \quad (4)$$

The first line is the closed-form solution to a weighted squared-residual objective with an ℓ_2 penalty. Here $\mathbf{D}_t$ is formed from current

input-window volatility, and the denominator shrinks low-support reads toward zero. When several events share an origin, NSRC averages their readouts equally, $\mathbf{a}_t = |\mathcal{E}_t|^{-1} \sum_{e \in \mathcal{E}_t} \mathbf{a}_t(e)$; an empty event set or memory gives $\mathbf{a}_t = \mathbf{0}$.

2.2. Correction strength from delayed feedback

The retrieved trajectory supplies a correction pattern, while completed historical outcomes set its amplitude. Let $\mathcal{B}_t \subseteq \mathcal{M}_t$ contain matured origins with nonzero readouts $\mathbf{a}_i$ computed at issuance; $n_t = |\mathcal{B}_t|$. At each update, NSRC fits a shared gain under both absolute and squared error:

$$g_{p,t} \in \arg \min_{g \in \mathbb{R}} \sum_{i \in \mathcal{B}_t} \omega_i \|\delta_i - g \mathbf{a}_i\|_p^p, \quad p \in \{1, 2\}. \quad (5)$$

Weights $\omega_i = (|\mathcal{S}_t| n_{s(i)} H)^{-1}$ give equal total weight to each represented asset and horizon; $\mathcal{S}_t$ is the asset set in $\mathcal{B}_t$, and $n_{s(i)}$ counts its records from asset $s(i)$. For $n_t \geq 3$,

$$g_t = \frac{n_t}{n_t + 10} [\min\{g_{1,t}, g_{2,t}, 1.5\}]_+, \quad \hat{\mathbf{u}}_t^s = \hat{\mathbf{u}}_t^b + g_t \mathbf{a}_t, \quad (6)$$

where $[x]_+ = \max(x, 0)$; otherwise $g_t = 0$. The gain is the smaller nonnegative estimate from the two losses, capped at 1.5 and shrunk when few completed examples are available. This yields the semantic correction $\hat{\mathbf{u}}_t^s$. NSRC also constructs a complementary uniform-relevance correction $\hat{\mathbf{u}}_t^a$ from the same archive, transfer, and calibration: it sets the event-match score to one for every record while retaining recency and asset weights. Both are internal candidate paths of NSRC.

2.3. Validation-frozen prediction routing

NSRC's router compares its semantic and uniform-relevance correction paths with the frozen backbone and persistence anchor. Be-

Table 1. Main forecasting results on four event cohorts (24-step pooled averages). Metrics are MSE ($\times 10^3$), MAE ($\times 10^2$), and MAPE (%). Bold/underline mark the best overall/learned results. NSRC uses validation-frozen routing; CAMEF uses event fusion, while other baselines use price history only.

Method	Celebrity events BTC/ETH/SOL/DOGE			News extension BTC/ETH/SOL/DOGE			Macro crypto BTC/ETH/SOL/DOGE			Macro equities SPX/NDX/INDU		
	MSE	MAE	MAPE	MSE	MAE	MAPE	MSE	MAE	MAPE	MSE	MAE	MAPE
Persistence [13]	0.77052	1.69004	1.66709	0.29355	1.18851	1.17952	0.83228	2.07600	2.08960	0.12917	0.83780	0.84204
DLinear [14]	0.91826	1.90757	1.87894	0.41033	1.43537	1.42205	0.90287	2.19579	2.21258	0.20139	1.14871	1.15083
GPT4TS [15]	0.94757	1.86991	1.85010	0.30436	1.19922	1.18919	0.85595	2.10147	2.11765	0.16474	0.97754	0.98286
PatchTST [16]	1.35421	2.21099	2.18146	0.39814	1.35129	1.33804	1.09775	2.40025	2.41861	0.30944	1.27346	1.27916
iTransformer [17]	1.35719	2.20417	2.17109	0.38716	1.36651	1.35332	1.24934	2.52978	2.55176	0.27950	1.24577	1.24917
TimesNet [18]	1.28604	2.18051	2.14036	0.43382	1.39458	1.38109	1.49577	2.78680	2.81560	0.30640	1.33303	1.33807
Time-LLM [19]	0.97559	1.87861	1.84509	0.31568	1.18681	1.17458	0.94044	2.25352	2.26913	0.15999	0.98954	0.99334
CAMEF [7]	0.84006	1.84039	1.81391	0.36687	1.38901	1.37757	0.83987	2.09482	2.10655	0.31838	1.51249	1.50798
Time-MoE [20]	1.46827	2.29085	2.25199	0.42549	1.38411	1.37112	1.33620	2.69795	2.72397	0.17004	1.00844	1.01245
NSRC (ours)	0.77448	1.68974	1.66632	0.29310	1.18509	1.17633	0.83228	2.07600	2.08960	0.12917	0.83780	0.84204

fore each test interval, validation selects one cohort-level path from these candidates and fixed persistence blends. With $\mathbf{p} = \mathbf{1}_H$ and $\mathcal{A} = \{.10, .25, .50, .75\}$, the candidate set is

$$\mathcal{C}_t = \{\mathbf{p}, \hat{\mathbf{u}}_t^b, \hat{\mathbf{u}}_t^a, \hat{\mathbf{u}}_t^s\} \cup \{(1 - \alpha)\mathbf{p} + \alpha\mathbf{q} : \mathbf{q} \in \{\hat{\mathbf{u}}_t^a, \hat{\mathbf{u}}_t^s\}, \alpha \in \mathcal{A}\}. \quad (7)$$

Validation $\mathcal{V}$ uses the preceding 30 days and only horizons completed before testing. Averaging over origins, horizons, and seeds gives

$$k^* = \arg \min_k^{\text{lex}} (\overline{\text{MAE}}_{\mathcal{V},k}, \overline{\text{MSE}}_{\mathcal{V},k}, k), \quad \hat{\mathbf{u}}_t^{\text{NSRC}} = \hat{\mathbf{u}}_t^{k^*}, \quad (8)$$

where MAE is primary, MSE breaks MAE ties, and candidate order breaks any remaining ties. The selected route stays fixed during testing, while memory and calibration update as earlier forecast origins fully mature. Outside event origins, NSRC retains the backbone forecast.

3. EXPERIMENTS

3.1. Data and protocol

The four cohorts are Celebrity (BTC/ETH/SOL/DOGE; train Jan–Feb 2025, test May), News (same assets; train Jan–Feb, test Jun–Aug), Macro crypto (same assets), and Macro equities (SPX/NDX/INDU); both macro cohorts train through August and test in November–December. They contain 288/61/164/117 event origins; News has no eligible DOGE event and one SOL event. The 30 days before each test interval select its route using only outcomes mature by test start. Because these intervals also informed method exploration, we report retrospective within-period forecasts.

Baseline citations appear in Table 1. DLinear, GPT4TS, and PatchTST use three seeds per cohort; iTransformer and TimesNet use three for Celebrity/News and one for macro; the remaining baselines use one. Price-only baselines share a 96-observation input, while Time-LLM, CAMEF, and Time-MoE use their official settings. NSRC uses DLinear for its forecast and residual branch, and freezes the validation-selected route per test interval.

For cohort D with N_D event origins, metrics pool all 24 horizons: $\langle f \rangle_D = (24N_D)^{-1} \sum_{i=1}^{N_D} \sum_{h=1}^{24} f_{i,h}$. Table 1 reports MSE=

$\langle \delta^2 \rangle_D$, MAE= $\langle |\delta| \rangle_D$, and MAPE= $100 \langle |\delta| / \max(|y|, 10^{-8}) \rangle_D$ for origin-price-normalized $y, \hat{y}$ and $\delta = y - \hat{y}$. Significance tests average seed-level loss across 24 horizons per event. Paired event differences are resampled in calendar-day blocks. All model decisions use pre-forecast information; 3-/14-day blocks test sensitivity.

3.2. Forecasting effectiveness

NSRC improves all 12 cohort–metric results over fixed DLinear and leads learned baselines in 11/12. Validation selects persistence for both macro cohorts and $0.9\mathbf{p} + 0.1\hat{\mathbf{u}}_t^a$ for Celebrity and News, where $\hat{\mathbf{u}}_t^a$ is the DLinear forecast with uniformly weighted residual memory. The gain comes from routing: a strong persistence anchor limits transfer when event-conditioned updates do not validate, while a small residual-path contribution helps in the other cohorts. These end-to-end results do not by themselves isolate semantic retrieval; Table 2 tests that mechanism separately.

The identical Macro entries reflect the selected Persistence route: these gains over DLinear come from the anchor, not a residual correction. For Celebrity and News, the selected 90/10 blend makes the calibrated correction a small adjustment to that anchor.

3.3. Residual-memory mechanism

Rows 2–13 of Table 2 analyze the unrouted semantic branch, while row 1 reports the deployed system. Relative to uniform relevance, semantic weighting improves all three errors for Celebrity and Macro equities, but not for News or Macro crypto; the utility of event addresses therefore depends on the cohort. The main router does not select this semantic path in the reported test intervals, so rows 2–13 compare memory mechanisms under a fixed unrouted path rather than explain deployed gains. Removing category has no effect for Celebrity/News because each has one test-time category.

Online updates and volatility scaling matter most in News, where newly matured residuals and current volatility help track a changing event mix. Delayed-feedback calibration helps News and Macro crypto; recency weighting helps Celebrity, News, and equities; cross-asset sharing is most useful in News and Macro crypto. Support shrinkage improves the semantic branch in all cohorts by limiting weakly supported reads. These differing effects explain why a single

Table 2. NSRC ablations (24-step means, seeds 2026–2028). Row 1 is the routed result; row 2 is the full unrouted DLinear semantic branch, and rows 3–13 remove one component from it. Metrics: MSE ($\times 10^3$), MAE ($\times 10^2$), MAPE (%). Category c is constant for Celebrity/News; fallback was inactive. Shuffled keys average 20 permutations.

Variant	Celebrity events			News extension			Macro crypto			Macro equities		
	MSE	MAE	MAPE	MSE	MAE	MAPE	MSE	MAE	MAPE	MSE	MAE	MAPE
NSRC (ours, routed)	0.77448	1.68974	1.66632	0.29310	1.18509	1.17633	0.83228	2.07600	2.08960	0.12917	0.83780	0.84204
Unrouted semantic branch	0.89821	1.86660	1.84003	0.31864	1.21778	1.21029	0.82462	2.08995	2.10129	0.15562	0.99349	0.99764
w/o semantic weighting	0.90709	1.88379	1.85628	0.31352	1.20989	1.20314	0.81066	2.06531	2.07642	0.16893	1.04461	1.04875
w/o description z	0.90498	1.87958	1.85228	0.31606	1.21335	1.20635	0.81799	2.07905	2.09015	0.16359	1.02122	1.02548
w/o category c	0.89821	1.86660	1.84003	0.31864	1.21778	1.21029	0.81409	2.07214	2.08324	0.16616	1.03208	1.03634
w/o direction d	0.90119	1.87232	1.84542	0.31441	1.21147	1.20432	0.81690	2.07697	2.08791	0.16965	1.04498	1.04923
w/o calibration	0.86242	1.79154	1.77472	0.32722	1.23380	1.22536	0.83179	2.10080	2.11231	0.15488	1.00568	1.00923
w/o online updates	0.91135	1.89456	1.86618	0.41033	1.43537	1.42205	0.81952	2.08602	2.09827	0.15537	0.97056	0.97577
w/o recency	0.89967	1.86921	1.84210	0.32517	1.24002	1.23259	0.81995	2.08303	2.09436	0.16382	1.00429	1.00934
w/o volatility scaling	0.87939	1.81950	1.79417	0.40520	1.43500	1.42266	0.83667	2.09879	2.10975	0.15463	0.99788	1.00169
w/o cross-asset sharing	0.88600	1.83495	1.80951	0.39460	1.41085	1.39919	0.83121	2.09356	2.10622	0.15504	0.99888	1.00266
w/o support shrinkage	0.91826	1.90757	1.87894	0.32923	1.25648	1.25201	0.82697	2.09550	2.10749	0.16201	1.00908	1.01361
shuffled keys (20×3)	0.89794	1.86601	1.83947	0.31760	1.21638	1.20900	0.81288	2.06408	2.07543	0.16535	1.03512	1.03897

Table 3. NSRC vs. DLinear. Δ = DLinear – NSRC in Table 1 units; seed-averaged 24-step paired origin losses use 49,999 circular 7-day bootstrap replicates. Brackets: unadjusted 95% CIs; p_H : Holm-adjusted two-sided p over 12 tests. Positive Δ favors NSRC.

Dataset	Metric	Δ [95% CI]	p_H
Celebrity	MSE	0.144 [0.017,0.307]	0.194
	MAE	0.218 [0.109,0.335]	0.0016
	MAPE	0.213 [0.106,0.325]	0.0008
News	MSE	0.117 [0.065,0.177]	0.0021
	MAE	0.250 [0.170,0.336]	0.00024
	MAPE	0.246 [0.166,0.330]	0.00024
Macro crypto	MSE	0.071 [-0.033,0.159]	0.361
	MAE	0.120 [-0.029,0.274]	0.361
	MAPE	0.123 [-0.029,0.278]	0.361
Macro equities	MSE	0.072 [0.029,0.118]	0.0090
	MAE	0.311 [0.129,0.510]	0.0091
	MAPE	0.309 [0.126,0.508]	0.0091

Table 4. Stability vs. DLinear (Table 1 units): Δ =pooled MSE/MAE/MAPE gains; H^+ =horizon wins (MSE/MAE/MAPE); E/L=early/late test-half MAE gains; V=low/mid/high-volatility MAE gains (pre-test tertiles); —=empty bin.

Dataset	Stability summary
Celebrity	Δ .144/.218/.213; H^+ 24/24/24
	E/L .264/.175; V .262/.465/-.101
News	Δ .117/.250/.246; H^+ 24/24/24
	E/L .275/.222; V .261/.164/—
Macro crypto	Δ .071/.120/.123; H^+ 20/17/17
	E/L .001/.213; V .202/.154/.040
Macro equities	Δ .072/.311/.309; H^+ 24/24/24
	E/L .119/.459; V .231/.301/.415

fixed transfer policy is inadequate. The fallback was inactive, so its

benefit is not measured here. Shuffled keys show no uniform degradation, consistent with cohort-dependent value from event addressing.

3.4. Reliability and stability

The paired tests compare the deployed route with fixed DLinear: Celebrity MAE/MAPE, all News metrics, and all Macro-equities metrics remain significant after Holm correction, while Macro crypto’s positive mean gains remain uncertain because all three CIs include zero. The 3- and 14-day block checks preserve this pattern. Table 4 shows positive MAE gains in both test halves for every cohort and gains across all horizons in three cohorts; Macro crypto retains gains at 17–20 of 24 horizons. The route loses in Celebrity’s high-volatility bin, marking large shocks as a boundary for the current correction scheme.

Figure 2 shows origin-normalized 24-step paths; the flat macro outputs follow from the selected Persistence route.

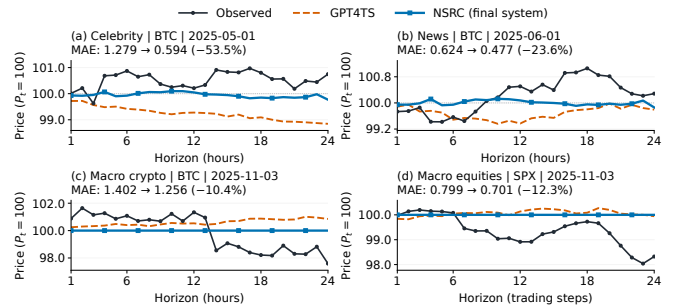


Fig. 2. Earliest-eligible BTC/SPX cases: GPT4TS vs. routed DLinear-based NSRC; titles show 24-step MAE. Chronological selection is independent of error.

4. CONCLUSION

Across four cohorts, NSRC’s event-addressed memory and validation-frozen routing improve all 12 cohort–metric results over fixed DLinear.

5. REFERENCES

- [1] A Craig MacKinlay, “Event studies in economics and finance,” *Journal of Economic Literature*, vol. 35, no. 1, pp. 13–39, 1997.
- [2] Werner Antweiler and Murray Z. Frank, “Is all that talk just noise? the information content of internet stock message boards,” *The Journal of Finance*, vol. 59, no. 3, pp. 1259–1294, 2004.
- [3] Nuno Oliveira, Paulo Cortez, and Nelson Areal, “The impact of microblogging data for stock market prediction: Using twitter to predict returns, volatility, trading volume and survey sentiment indices,” *Expert Systems with Applications*, vol. 73, pp. 125–144, 2017.
- [4] Robert J Shiller, *Narrative economics: How stories go viral and drive major economic events*, Princeton University Press, 2019.
- [5] S. A. Farimani et al., “An adaptive multimodal learning model for financial market price prediction,” *IEEE Access*, vol. 12, pp. 121847–121858, 2024.
- [6] Siyi Zhang, Tong Che, Zhiliang Zhu, et al., “Forecasting of exchange rate time series based on event-aware transformer model,” *Soft Computing*, vol. 29, pp. 3035–3045, 2025.
- [7] Yang Zhang, Wenbo Yang, Jun Wang, Qiang Ma, and Jie Xiong, “Camef: Causal-augmented multi-modality event-driven financial forecasting by integrating time series patterns and salient macroeconomic announcements,” in *Proceedings of the 31st ACM SIGKDD Conference on Knowledge Discovery and Data Mining*, 2025, pp. 3867–3878.
- [8] Changsen Wang, Qi Qi, Jingyu Wang, Haifeng Sun, Zirui Zhuang, and Jianxin Liao, “Chattime: A unified multimodal time series foundation model bridging numerical and textual data,” in *Proceedings of the AAAI Conference on Artificial Intelligence*, 2025, vol. 39.
- [9] Chenxi Liu, Zhu Xiao, Cheng Long, Dong Wang, Tao Li, and Hongbo Jiang, “Timecma: Towards llm-empowered multivariate time series forecasting via cross-modality alignment,” in *Proceedings of the AAAI Conference on Artificial Intelligence*, 2025, vol. 39.
- [10] Yuchen He, Yueyang Cang, Zhiyuan Ning, Ningyu Wang, and Li Shi, “Ratl: Learning from retrieved residuals for robust multivariate time-series forecasting,” 2026.
- [11] Robin Manhaeve, Sebastijan Dumancic, Angelika Kimmig, Thomas Demeester, and Luc De Raedt, “Deepproblog: Neural probabilistic logic programming,” in *Advances in Neural Information Processing Systems*, S. Bengio, H. Wallach, H. Larochelle, K. Grauman, N. Cesa-Bianchi, and R. Garnett, Eds. 2018, vol. 31, Curran Associates, Inc.
- [12] Jiayuan Mao, Chuang Gan, Pushmeet Kohli, Joshua B. Tenenbaum, and Jiajun Wu, “The Neuro-Symbolic Concept Learner: Interpreting Scenes, Words, and Sentences From Natural Supervision,” in *International Conference on Learning Representations*, 2019.
- [13] Rob J. Hyndman and George Athanasopoulos, *Forecasting: Principles and Practice*, OTexts, 3 edition, 2021.
- [14] Ailing Zeng, Muxi Chen, Lei Zhang, and Qiang Xu, “Are transformers effective for time series forecasting?,” in *Proceedings of the AAAI Conference on Artificial Intelligence*, 2023, vol. 37, pp. 11121–11128.
- [15] Tian Zhou, Peisong Niu, Liang Wang, Jian Sun, and Rong Jin, “One fits all: Power general time series analysis by pretrained lm,” in *Advances in Neural Information Processing Systems*, 2023, vol. 36, pp. 43322–43355.
- [16] Yuqi Nie, Nam H Nguyen, Phanwadee Sinthong, and Jayant Kalagnanam, “A time series is worth 64 words: Long-term forecasting with transformers,” in *International Conference on Learning Representations (ICLR)*, 2023.
- [17] Yong Liu, Tengge Hu, Haoran Zhang, Haixu Wu, Shiyu Wang, Lintao Ma, and Mingsheng Long, “itransformer: Inverted transformers are effective for time series forecasting,” in *International Conference on Learning Representations (ICLR)*, 2024.
- [18] Haixu Wu, Tengge Hu, Yong Liu, Hang Zhou, Jianmin Wang, and Mingsheng Long, “Timesnet: Temporal 2d-variation modeling for general time series analysis,” in *International Conference on Learning Representations*, 2023.
- [19] Ming Jin, Shiyu Wang, Lintao Ma, Zhixuan Chu, James Y Zhang, Xiaoming Shi, Pin-Yu Chen, Yuxiao Liang, Yuan-Fang Li, Shirui Pan, and Qingsong Wen, “Time-llm: Time series forecasting by reprogramming large language models,” in *International Conference on Learning Representations (ICLR)*, 2024.
- [20] Xiaoming Shi, Shiyu Wang, Yuqi Nie, Dianqi Li, Zhou Ye, Qingsong Wen, and Ming Jin, “Time-moe: Billion-scale time series foundation models with mixture of experts,” 2024.